AYUSH BARIK

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RESEARCH INTERESTS

Diffusion/Flow-Matching, Computer Vision, Audio, Algorithms, Inference, Systems for ML, Efficiency

EDUCATION

University of Illinois Urbana-Champaign *Aug. 2023 – May 2027*
Bachelor of Science in Computer Engineering, GPA: 3.90/4.00 *Illinois, USA*
James Scholar Honors Student
Advised By Prof. Fan Lai, Prof. Francis Yan, and Prof. Romit Roy Choudhury

PUBLICATIONS & PREPRINTS

1. **SoundWeaver: Compositional Warm-Starting for Text-to-Audio Diffusion Serving**
Ayush Barik[1], Sofia Stoica[1], Nikhil Sarda[2], Arnav Kethana[1], Abhinav Khanduja[1], Muchen Xu[1], Fan Lai[1]
[1]UIUC, [2]Assured Intelligence
(ICLR 2027 in-submission)

2. **VideoRAG+: Rethinking Graph-Structured Retrieval for Long Video Understanding**
Ayush Barik[1], Fan Lai[1], Francis Yan[1]
[1]UIUC
(CVPR 2027 submitting-soon)

3. **CHOIR: Generating High-Fidelity Room Impulse Responses Using Conditional Flows**
Chaitanya Amballa[1], Ayush Barik[1], Steve Lin[1], Jim Sun[2], Mehmet Ergezer[2], Vamshi Krishna Chillara[2], Wontak Kim[2], Romit Roy Choudhury[1,2]
[1]UIUC, [2]Amazon
(ICASSP 2027 in-submission)

4. **FlashDiff: Efficient Regional Execution and Scheduling for Diffusion Model Serving**
Yaqi Qiao[1], Ping He[2], Songrun Xie[3], Ayush Barik[1], Chensong Zhang[4], Zhengzhong Tu[5], Fan Lai[1]
[1]UIUC, [2]Vanderbilt University, [3]HKUST, [4]Nvidia, [5]Texas A&M University
(MLSys 2027 submitting-soon)

5. **RewardFlow: Generate Images by Optimizing What You Reward**
Onkar Susladkar[1], Dong-Hwan Jang[1], Tushar Prakash[2], Adheesh Juvekar[1], Vedant Shah[1], Ayush Barik[1], Nabeel Bashir[1], Muntasir Wahed[1], Ritish Shrirao[1], Ismini Lourentzou[1]
[1]UIUC, [2]Sony
(CVPR 2026)

6. **Toward Engineering AGI: Benchmarking the Engineering Design Capabilities of LLMs**
Xingang Guo[1], Yaxin Li[1], Xiangyi Kong[1], . . . , Ayush Barik[1], . . . , Bin Hu[1]
[1]UIUC, [2]University of Pennsylvania, [3]UC San Diego, [4]University of Michigan, [5]Amazon AGI
(NeurIPS 2025)

PROJECTS

GPT-2 Inference Engine (CUDA)
Built GPT-2 Inference from scratch (attention, FFN, LayerNorm) and profiled on NVIDIA A40 GPUs; optimized with FlashAttention-2, KV caching, CUTLASS kernel fusion, double-buffering, and shared memory swizzling.

Real-Time Quantized Video Streaming on FPGA (SystemVerilog)
Real-time OV7670 camera feed rendered to HDMI from on-chip BRAM. SCCB controller configures the camera to output a quantized 12-bit RGB444 Quantized VGA stream. Everything composited in 813 LUTs at 0.35 W.

TEACHING

ECE 220: Computer Systems & Programming — *Fall 2024, Spring 2025, Fall 2025*
Undergrad TA, University of Illinois Urbana-Champaign — *Illinois, USA*

MATH 257: Linear Algebra w/ Computational Applications — *Summer 2025*
Undergrad TA, University of Illinois Urbana-Champaign — *Illinois, USA*

ECE 364: Programming Methods for Machine Learning — *Fall 2025, Fall 2026*
Undergrad TA, University of Illinois Urbana-Champaign — *Illinois, USA*

USA Computing Olympiad Tutor — *Fall 2024*
Private Online Tutor — *Online*

English Tutor — *Spring 2022, Fall 2022*
DMNAPK Rehabilitation Center for Young Girls — *Dhaka, Bangladesh*

SERVICE

Reviewer
ICLR (2027)

AWARDS

Illinois Engineering Achievement Scholarship — *Fall 2024*